Final exam – practice

Write a colab notebook, *clearly indicating problem section like 1(d)*, for problems 1(a-e) and 2(d-e). Answer problem 2(a-c) on paper and take photo. Upload both the .ipynb file and the .jpg or .pdf files onto canvas. Available time: 120 minutes.

You may use only: your private notes, the canvas page of the course, and documentation of the `numpy` library. Using generative AI is not permitted; AI assistance in colab should be turned off. (If generative AI use is suspected, the student will be asked to prove full understanding of every line of code in the submitted notebook.) **Using not permitted resources result not only in failing this course, but also in disciplinary action.**

Problem 1

Consider a dataset with input variables $\boldsymbol{x} \in \mathbb{R}^5$ and target variables $t \in \mathbb{R}$; your task will be to predict the target for unseen $\boldsymbol{x}$ values. The training dataset (with 6 columns: the 5 components of $\boldsymbol{x}$ and the single value of t, as indicated in the header) is available as `train.csv`, and the test dataset (with 5 columns: only the components of $\boldsymbol{x}$) is available as `test.csv`.

You may use numpy (and pandas if desired), but *not* scikit-learn.

a) Load the data files, verify their data shape.

b) Reduce the dimension of the input $\boldsymbol{x}$ from 5 to 1. Specify the name of the algorithm used.

c) Plot t vs. x_1 ($\boldsymbol{x}$ reduced to 1 dimension). Decide what would be the most suitable degree for a polynomial fit (eg. linear or quadratic or ...).

d) Perform polynomial regression (of the degree you just selected) of t vs. x_1 on the training data. Plot both t and its predicted value vs. x_1. Determine what is the expected uncertainty of the predictions.

e) Using the above procedure, predict for the test dataset. Print the predicted values.

Problem 2

Consider the distribution X with the following probability density function:

$$p(x) = A \exp\left(-\frac{\sqrt{2}|x-\mu|}{\lambda}\right)$$

for $x \in [-\infty, \infty]$, where μ and λ are parameters.

a) Determine A, such that $p(x)$ is properly normalized.

b) What is the mean of this distribution?

c) Show, by integration by parts, that the variance of the distribution is λ^2.

d) Sample numerically the distribution, e.g., using `np.random.default_rng()`:
Generate $u_1, u_2 \sim \text{Uniform}[0,1]$.
Set $s = \begin{cases} 1 & \text{if } u_1 < 0.5 \\ -1 & \text{otherwise} \end{cases}$
Generate the sample as $x = s\lambda \log(u_2)/\sqrt{2}$.
For $\lambda = 3$, show that the histogram of x (e.g., using `matplotlib.pyplot.hist`) matches the probability density function.

e) Demonstrate numerically that for three different values of λ the variance of X is indeed λ^2.